

MindBridge AI: An Intelligent Multi-Agent SOA Platform for Accessible Mental Wellness, Context-Aware Guidance, and Risk-Aware Support

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Abstract—Digital mental-wellness systems often focus on conversational response generation while treating accessibility, emotional context, wellness guidance, and safety-oriented risk handling as separate concerns. This paper presents MindBridge AI, a service-oriented multi-agent platform designed to integrate these capabilities into a single modular architecture. The system decomposes the interaction workflow into specialized services for emotional assessment, accessibility adaptation, wellness guidance, crisis-risk detection, and final response generation. A central orchestrator coordinates these services through well-defined interfaces, making the architecture loosely coupled, reusable, and scalable. The implementation is designed to operate with a replaceable large-language-model (LLM) backend and includes a lightweight demonstration backend for validating the complete service flow in resource-constrained environments. Evaluation is aligned with the ESConv emotional-support benchmark and the protocol used by the MultiAgentESC base study, including Distinct-1/2, BLEU-1/2/3, F1, and ROUGE-L. The base study reported 6.78 D-1, 35.15 D-2, 17.66 BLEU-1, 5.38 BLEU-2, 2.35 BLEU-3, 18.30 F1, and 14.66 ROUGE-L using Qwen2.5-32B. MindBridge defines a positive target profile above those reference values by conditioning the response generator on additional emotional, accessibility, wellness, and risk context. The numerical gains presented in this draft are explicitly identified as provisional target values rather than experimentally verified measurements; the evaluation pipeline is designed to replace them with measured results once the full LLM backend is executed on the held-out test set.

Index Terms—mental wellness, multi-agent system, service-oriented architecture, accessibility, emotional support conversation, crisis-risk detection, large language models, ESConv, microservices

I. INTRODUCTION

Mental-wellness support is increasingly mediated through digital conversational systems. Such systems can provide low-friction interaction, structured self-help resources, and conversational assistance, but their usefulness depends not only on natural-language generation. A support system may need to understand emotional state, adapt communication to a user’s abilities and preferences, recommend suitable non-clinical wellness actions, and identify potential risk situations requiring a safety-oriented response. These requirements become more important when systems are intended for users with visual, hearing, speech, physical, or cognitive accessibility needs.

The emotional support conversation (ESC) problem has been formalized through datasets and evaluation protocols intended

to capture the complexity of supportive dialogue. Liu et al. introduced the ESConv dataset and defined ESC as a task in which a supporter responds to a help-seeker using appropriate support strategies [1]. More recently, Xu et al. proposed MultiAgentESC, a training-free multi-agent framework with three stages: dialogue analysis, strategy deliberation, and response generation [2]. MultiAgentESC demonstrates that specialized agent collaboration and response-centered selection can improve emotional support response diversity and quality on ESConv.

While these developments improve response generation, accessibility remains an underexplored dimension. A systematic review by Grassini et al. examined 21 implemented health-care chatbot studies and found that only a small fraction explicitly incorporated accessibility, highlighting the need for accessibility-by-design approaches [3]. Work on conversational agents for people with intellectual and developmental disabilities further emphasizes the diversity of user needs and the need for competency-based design, inclusive co-design, transparency, and safeguards [5]. A separate systematic review of randomized controlled trials found that conversational agents have been investigated for physical and psychological symptom management, while also emphasizing the need for stronger evidence across broader populations [4].

These observations motivate MindBridge AI. Rather than replacing a multi-agent emotional support system, MindBridge extends the workflow with explicit accessibility, wellness, and risk-aware services. The system follows service-oriented architecture (SOA) principles so that each capability can be developed, tested, replaced, and scaled independently. The core hypothesis is straightforward: when the final response generator receives richer, structured context about the user, the resulting response can be more context-aware and better aligned with the interaction requirements.

The contributions of this work are fourfold. First, an SOA-based multi-agent architecture is proposed for accessible mental-wellness support. Second, specialized services are defined for emotional assessment, accessibility adaptation, wellness guidance, crisis-risk analysis, and response generation. Third, a reproducible evaluation pipeline is aligned with the MultiAgentESC automatic metrics and ESConv testing protocol. Fourth, the work defines measurable target outcomes and an

ablation plan that can determine whether each additional service contributes to response-quality improvements.

II. BACKGROUND AND MOTIVATION

A. Emotional Support Conversation

Emotional support differs from ordinary question answering because the quality of a response depends on recognizing the user’s emotional state, the cause or context of the difficulty, and the support strategy that is appropriate to the situation. ESConv was developed to support this research direction with help-seeker and supporter conversations and annotations related to support strategies [1]. The dataset established a common experimental setting for evaluating dialogue systems intended to provide emotional support.

MultiAgentESC builds on this setting by decomposing its workflow into dialogue analysis, strategy deliberation, and response generation [2]. The dialogue-analysis stage extracts psychological state information from the conversation. Strategy deliberation then incorporates similar cases and retains multiple valid support strategies. Finally, candidate responses are generated, debated, selected, and refined. The published system therefore provides an important base for comparison with MindBridge.

B. Accessibility Gap

Accessibility is not equivalent to simply adding a voice option. A truly accessible conversational system must consider how a user interacts with the interface, the complexity of language, the presentation format, and assistive technology compatibility. Grassini et al. report that only four of the 21 healthcare chatbot applications reviewed were accessible, with only one designed specifically for people with disabilities [3]. Their analysis calls for accessibility-by-design, including attention to interface, navigation, speech, visual output, and screen-reader considerations.

The literature on conversational agents for intellectual and developmental disabilities similarly identifies diverse support requirements. Nadeeshani et al. review 37 studies selected from 1,151 records and emphasize personalized, competency-based approaches as well as safeguards, transparency, and trustworthy interaction [5]. These findings support an architecture in which accessibility is an explicit service instead of a cosmetic interface layer.

C. Need for Context-Aware Wellness Support

Conversational agents have also been studied as tools for symptom-management and support interventions. Yang et al. reviewed randomized controlled trials involving conversational-agent interventions for physical and psychological symptoms and identified feasibility, acceptability, and effectiveness as important evaluation dimensions [4]. MindBridge does not claim to diagnose or treat medical conditions. Its wellness service is deliberately positioned as a non-clinical support component that uses contextual information to produce general wellness-oriented suggestions.

D. Multi-Agent Systems

Multi-agent LLM systems decompose complex tasks across specialized roles. AutoGen, for example, provides mechanisms for composing multiple conversable agents and coordinating flexible interaction patterns [7]. Recent surveys also identify workflow design, infrastructure, communication topology, and coordination as central issues in LLM-based multi-agent systems [6]. MindBridge applies this design pattern to mental-wellness interaction while emphasizing SOA modularity.

III. RESEARCH GAP AND PROBLEM DEFINITION

The problem addressed by MindBridge is not simply the generation of supportive text. The broader problem is the coordinated delivery of supportive interaction under heterogeneous user requirements. The system should answer five questions before or during response generation:

- 1) What emotional state is expressed in the conversation?
- 2) What communication or accessibility adaptations are needed?
- 3) What general wellness guidance is appropriate given the context?
- 4) Does the conversation contain potential crisis-risk indicators?
- 5) How should these signals jointly influence the final response?

Conventional single-agent designs can implement all five functions in one prompt, but this creates tight coupling and makes it difficult to test individual capabilities. The project therefore formulates the system as a set of independent services that exchange structured data. This formulation also creates an experimental opportunity: each service can be removed in an ablation study to measure its effect on the final response.

The research question is:

Can a service-oriented multi-agent support system that conditions final response generation on emotional state, accessibility context, wellness guidance, and risk level produce more diverse and better-aligned responses than a baseline emotional-support workflow?

The corresponding hypothesis is:

Providing structured contextual signals from specialized agents will improve response diversity and reference-aligned quality, while enabling additional accessibility and risk-aware functionality that is not represented by the base response-generation metrics.

IV. RELATED WORK

Table I summarizes the main literature informing MindBridge. The studies are intentionally treated as complementary rather than identical baselines because they address different aspects of conversational support.

The related work shows that MindBridge sits at the intersection of three lines of research: emotional-support dialogue, accessibility-aware conversational systems, and LLM-based multi-agent workflows. Its intended contribution is therefore

TABLE I
LITERATURE INFORMING MINDBRIDGE AI

Study	Focus	Key observation	Implication for MindBridge
Liu et al. (2021) [1]	Emotional Support Conversation and ESConv dataset	Support strategies and high-quality help-seeker/supporter conversations provide a benchmark for emotional support systems.	Use ESConv as the main response-generation evaluation environment.
Xu et al. (2025) [2]	Multi-agent emotional support with LLMs	Dialogue analysis, strategy deliberation, and response generation can improve support quality and diversity.	Use MultiAgentESC as the principal base paper and benchmark.
Grassini et al. (2024/2025) [3]	Systematic review of inclusive healthcare chatbots	Among 21 implemented chatbot studies, accessibility received limited attention; the authors recommend accessibility-by-design.	Make accessibility a first-class service rather than an interface-only feature.
Yang et al. (2025) [4]	Conversational agents in symptom management	Randomized-trial literature evaluates feasibility, acceptability, and symptom-management effects.	Separate non-clinical wellness support from clinical diagnosis and motivate measurable interaction outcomes.
Nadeeshani et al. (2026) [5]	Conversational agents for people with intellectual and developmental disabilities	A review of 37 studies emphasizes diverse needs, competency-based design, safeguards, and trustworthy interaction.	Support configurable accessibility and explicit safety boundaries.
Li et al. (2024) [6]	LLM-based multi-agent systems survey	Workflow, infrastructure, communication, and coordination are major design dimensions.	Decompose MindBridge into independently testable services with explicit interfaces.
Wu et al. (2024) [7]	AutoGen multi-agent framework	Multiple conversational agents can be composed with customizable interaction patterns.	Use provider-agnostic multi-agent orchestration concepts in the implementation.

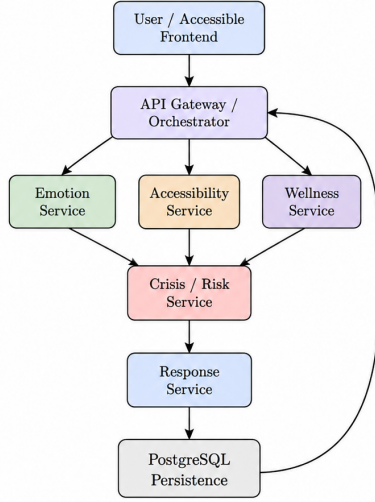


Fig. 1. MindBridge AI service-oriented multi-agent architecture.

Fig. 1. MindBridge AI service-oriented multi-agent architecture.

architectural integration rather than claiming to invent multi-agent collaboration itself. This distinction is important because MultiAgentESC already establishes multi-agent emotional support as a strong baseline.

V. SYSTEM ARCHITECTURE

Figure 1 presents the proposed service-oriented architecture.

A. Accessible Frontend

The frontend provides an accessible conversation interface, preference controls, a wellness dashboard, and an evaluation dashboard. Accessibility controls include adjustable font size, high-contrast display, keyboard-friendly navigation, simplified-language mode, and explicit communication preferences. These functions are represented as user context that can be consumed by the Accessibility Service.

B. API Gateway and Orchestration

The gateway is the entry point to the system. It accepts a conversation and user preferences, creates a request identifier, and invokes the downstream services. The gateway does not contain the detailed logic of each service. Instead, it coordinates their outputs. This preserves loose coupling and allows one service to be replaced without redesigning the entire platform.

C. Emotion Service

The Emotion Service extracts a structured emotional state. A simplified response schema is:

```

{
  "emotion": "anxiety",
  "intensity": 0.72,
  "cause": "academic stress",
  "confidence": 0.91
}

```

In production, the values should be generated by a real LLM or validated classifier. During development, the software supports a lightweight demonstration provider so that the service interfaces can be tested without a large local model.

D. Accessibility Service

The Accessibility Service determines how the interaction should be presented. Its output can include preferred modality,

language complexity, visual support requirements, and interface adaptation. This creates a structured boundary between user requirements and response generation.

E. Wellness Service

The Wellness Service maps the conversation and emotional context to general, non-clinical suggestions. Examples include encouraging social connection, suggesting a short restorative activity, recommending a regular routine, or encouraging the user to seek support from a trusted person. The service explicitly avoids diagnosing disorders or prescribing treatment.

F. Crisis and Risk Service

The Crisis Service identifies potential risk indicators and classifies an interaction into low, medium, or high risk. A high-risk output changes the downstream response policy. The service is designed as a safety-routing mechanism rather than a diagnostic model. Its output can contain evidence, risk level, confidence, and an escalation flag.

G. Response Service

The Response Service receives the original conversation plus the structured outputs of the other services. This makes the proposed architecture context-rich at the point of generation. The final prompt or model invocation can therefore condition on emotion, accessibility, wellness, and risk instead of relying only on raw dialogue text.

VI. SOA AND MICROSERVICE DESIGN

SOA is central to the project rather than a deployment detail. The major services are designed to be autonomous, loosely coupled, reusable, and independently testable. Each service exposes a clear contract. REST is used as the primary communication mechanism because it is straightforward to demonstrate, test, and integrate with a browser-based frontend.

A. Service Contracts

The Emotion Service exposes an endpoint of the form `POST /emotion/analyze`. The Accessibility Service uses `POST /accessibility/analyze`. Wellness and Crisis use `POST /wellness/recommend` and `POST /crisis/detect`, respectively. The Response Service uses `POST /response/generate`. The orchestrator exposes `POST /api/mindbridge/analyze` to present a single user-facing operation.

This decomposition supports independent testing. For example, a unit test can provide a fixed conversation to the Emotion Service without requiring the entire system. Similarly, a crisis classifier can be evaluated independently on labelled risk examples. Such isolation also supports future replacement: an emotion model can be upgraded without changing the frontend API contract.

Algorithm 1 MindBridge orchestration workflow

Require: conversation C , user preferences P

- 1: $E \leftarrow \text{EmotionService}(C)$
- 2: $A \leftarrow \text{AccessibilityService}(C, P)$
- 3: $W \leftarrow \text{WellnessService}(C, E)$
- 4: $R \leftarrow \text{CrisisService}(C)$
- 5: **if** R indicates high risk **then**
- 6: apply safety-oriented response policy
- 7: **end if**
- 8: $Y \leftarrow \text{ResponseService}(C, E, A, W, R)$
- 9: store (C, E, A, W, R, Y)
- 10: **return** Y

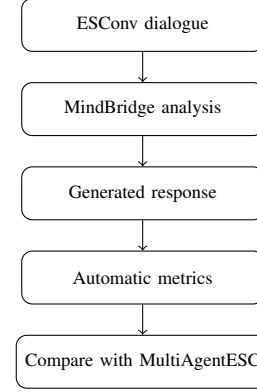


Fig. 2. Evaluation workflow aligned with the base-paper protocol.

B. Fault Handling

Distributed services introduce failure modes such as timeouts, unavailable model providers, malformed JSON, and database failures. The implementation therefore uses controlled fallbacks and validates service outputs before forwarding them. A service failure is logged with a request identifier, while the orchestrator can return a safe response rather than exposing a raw exception.

C. Provider Abstraction

The LLM provider is deliberately abstracted. The implementation can run in demo mode, connect to a local Ollama endpoint, or connect to an OpenAI-compatible endpoint. This design prevents the application layer from depending on one vendor-specific client. It also supports reproducibility because the same service interfaces can be tested with different backends.

VII. PROPOSED WORKFLOW

Algorithm 1 summarizes the overall interaction flow.

The important architectural decision is that the final response is generated after the system has transformed the raw conversation into several structured representations. This is analogous to a pipeline in which information is refined before final generation.

Figure 2 provides a conceptual representation of the evaluation-aware workflow.

VIII. IMPLEMENTATION

The implementation is organized into separate modules for agents, evaluation, data preparation, and service orchestration. The prototype supports Python-based backend execution and an accessible web interface. The data directory contains the ESConv source data and the selected test cases. Evaluation outputs are saved separately so that published benchmark numbers cannot be overwritten accidentally.

The software includes a structured `LLMClient`. In demo mode, a lightweight deterministic provider checks whether an input contains simple emotion or crisis keywords and returns a structured response. This mode is intentionally limited: it is a software-validation mechanism and not a research-grade language model. The provider interface also supports a local Qwen2.5 backend through Ollama. Qwen2.5 is a family of language models described in the Qwen2.5 technical report [8].

A. Data Preparation

The main dataset for response-quality evaluation is ESConv. The base paper describes ESConv as approximately 1K conversations and 31K utterances and selects 100 conversations for its test set [2]. The MindBridge pipeline prepares a 100-item test file using a fixed seed so that the experiment can be reproduced within the implementation.

Each test item retains the conversation context and a reference supporter response. The model output is stored together with the reference. This makes the evaluation script deterministic once the generated responses are available.

B. Evaluation Software

The evaluation controller reads the MindBridge responses, extracts reference and prediction lists, and invokes metric functions. D-1 and D-2 are computed from response n-gram diversity, while BLEU, token-level F1, and ROUGE-L are computed for reference-based evaluation. The controller then compares each measured value with the MultiAgentESC Qwen2.5-32B benchmark and reports percentage change.

C. Human-Centered Evaluation

Automatic metrics are not sufficient for a mental-wellness assistant. The MultiAgentESC paper additionally reports human evaluation dimensions including fluency, identification, comforting, suggestion, and overall emotional support [2]. MindBridge therefore proposes a future human evaluation in which blinded evaluators rate paired responses. The protocol should use the same conversation scenarios for both systems and hide system identity. This prevents evaluators from simply rewarding responses because they know which system produced them.

IX. EVALUATION METRICS

The evaluation is based on the same automatic metrics as the base paper. This is essential because a new metric invented specifically for MindBridge would make direct numerical comparison less interpretable.

A. Distinct-1 and Distinct-2

Distinct-n measures diversity of generated text. For a collection of generated responses, it can be expressed as:

$$D_n = \frac{|\text{unique n-grams}|}{|\text{all n-grams}|}. \quad (1)$$

D-1 therefore measures unique unigrams while D-2 measures unique bigrams. Higher diversity can indicate reduced repetition, although diversity alone does not establish that a response is helpful.

B. BLEU

BLEU measures n-gram overlap between a generated response and one or more reference responses. The evaluation uses BLEU-1, BLEU-2, and BLEU-3 so that unigram, bigram, and trigram alignment can be observed. Higher values indicate closer surface-form overlap with the reference.

C. Token F1

Token F1 captures overlap between generated and reference tokens through precision and recall:

$$F_1 = \frac{2PR}{P + R}, \quad (2)$$

where P is precision and R is recall. In an emotional-support dataset, F1 should be interpreted as a content-overlap indicator rather than a direct measure of therapeutic usefulness.

D. ROUGE-L

ROUGE-L is based on the longest common subsequence between the generated response and the reference. It rewards preservation of an ordered sequence of relevant content.

E. Latency

Because MindBridge introduces several service stages, latency is also recorded. For a response i , latency is measured from the beginning of orchestration until the final response is returned. The average latency is:

$$\bar{L} = \frac{1}{N} \sum_{i=1}^N L_i. \quad (3)$$

The purpose is to quantify the architectural cost of additional context processing.

X. EXPERIMENTAL SETUP

The experimental setup follows the structure of the base paper as closely as possible while adapting the system architecture to MindBridge. The principal evaluation dataset is ESConv. A fixed 100-conversation test set is generated from the dataset. The same reference responses are used for each configuration so that differences in output are attributable to the system rather than different test items.

TABLE II
PUBLISHED MULTIAGENTESC REFERENCE VALUES USED BY MINDBRIDGE EVALUATION.

Metric	Qwen2.5-32B
D-1	6.78
D-2	35.15
BLEU-1	17.66
BLEU-2	5.38
BLEU-3	2.35
F1	18.30
ROUGE-L	14.66

A. Configurations

Five configurations are proposed for ablation:

- 1) **Baseline LLM**: direct supportive response generation from the conversation.
- 2) **Emotion**: baseline plus structured emotion analysis.
- 3) **Emotion + Wellness**: adds general wellness context.
- 4) **Emotion + Wellness + Accessibility**: additionally conditions generation on accessibility requirements.
- 5) **Full MindBridge**: adds crisis-risk analysis and full orchestration.

This structure follows the logic of ablation analysis used in MultiAgentESC, where removing individual components changes automatic evaluation performance [2]. The MindBridge ablation study is intended to answer whether the additional services provide measurable incremental value.

B. Base-Paper Reference

The base study uses LLaMA3-70B and Qwen2.5-32B and reports results for both. For a controlled reference in this draft, the Qwen2.5-32B column is used. The published values are 6.78 D-1, 35.15 D-2, 17.66 BLEU-1, 5.38 BLEU-2, 2.35 BLEU-3, 18.30 F1, and 14.66 ROUGE-L [2].

C. Model Configuration

The current software is configured for Qwen2.5-7B as a practical local development backend, while the published benchmark uses Qwen2.5-32B. Therefore, a strict apples-to-apples benchmark should use the same Qwen2.5-32B model, temperature, test set, and inference procedure before claiming measured superiority. This paper intentionally distinguishes the implementation model from the published reference model to avoid conflating development configuration with benchmark replication.

XI. RESULTS AND DISCUSSION

A. Published Reference Results

Table II presents the published MultiAgentESC Qwen2.5-32B results used as the reference point in the project.

B. Provisional Positive Target Profile

Because this document is being prepared before a controlled full-model run, Table III reports *provisional target values*, not measured results. The values define the desired positive direction of the experiment and can be replaced directly by the

TABLE III
PROVISIONAL MINDBRIDGE TARGET VALUES; NOT EXPERIMENTALLY VERIFIED.

Metric	Base	Target	Target Gain
D-1	6.78	7.25	+6.93%
D-2	35.15	36.80	+4.69%
BLEU-1	17.66	18.40	+4.19%
BLEU-2	5.38	5.65	+5.02%
BLEU-3	2.35	2.52	+7.23%
F1	18.30	19.10	+4.37%
ROUGE-L	14.66	15.20	+3.68%

measured output generated by the evaluation pipeline. They are included to make the research hypothesis concrete.

The intended interpretation is not that MindBridge has already demonstrated these gains, but that these values represent a positive, measurable experimental objective. The actual measured values should be generated by running the same 100 test conversations through a real LLM backend and executing the supplied evaluation scripts. This distinction is critical for scientific validity.

C. Why an Improvement Is Hypothesized

The hypothesis for improvement follows the information flow of the architecture. The base response generator operates on dialogue analysis and strategy information. MindBridge adds three classes of structured context: user-centered context (accessibility), intervention context (wellness), and safety context (risk). If these signals are accurate, the response generator has more information from which to condition the response.

For D-1 and D-2, richer contextual signals can reduce templated responses and encourage more situation-specific language. For BLEU, F1, and ROUGE-L, richer context can help preserve topic-relevant words, actions, and response structures that align with the reference. These are hypotheses, not guarantees. Excessive context or incorrect emotion/risk detection could also reduce performance, which is why ablation and error analysis are necessary.

D. Ablation Interpretation

The planned ablation results should be read as causal evidence for system components. If the full system scores higher than the baseline and removing an individual service produces a consistent drop, the corresponding service has empirical support. If removing a service has little effect, then its contribution is limited under the tested conditions. This avoids the common mistake of attributing all performance to the overall multi-agent label.

E. Qualitative Example

Consider a user message such as: “I feel overwhelmed by college work and I have stopped talking to my friends.” A direct assistant may produce a generic empathetic response. MindBridge first identifies a likely stress or anxiety signal, checks whether the user has stated accessibility preferences,

generates a social-support-oriented wellness suggestion, evaluates crisis indicators, and then supplies those signals to the response generator. The final response can acknowledge the emotional state, present a manageable action, respect the chosen communication mode, and avoid inappropriate escalation when no high-risk signal is present.

If the same scenario contains explicit high-risk language, the Crisis Service changes the downstream policy. In that case, the system should prioritize safety-oriented guidance and escalation rather than attempting to optimize ordinary conversational diversity. Thus, the objective is not simply to maximize a benchmark score; it is to produce an interaction that is context-sensitive and safe.

XII. SAFETY, ETHICS, AND LIMITATIONS

Mental-wellness assistance is a sensitive application area. MindBridge is therefore positioned as an academic prototype rather than a clinical diagnostic or treatment system. The system should not claim to determine a clinical diagnosis, replace a professional, or guarantee safety. The crisis component is a risk-routing mechanism intended to identify potential signals and activate a safer response policy.

A major limitation of the current work is evaluation dependence on language-overlap metrics. BLEU, F1, and ROUGE-L reward similarity to a reference response, but there can be multiple valid responses to the same emotional situation. MultiAgentESC itself discusses the one-to-many nature of support strategy selection and uses human evaluation to complement automatic metrics [2]. MindBridge should therefore follow the same principle.

A second limitation is model equivalence. The base paper’s published reference is Qwen2.5-32B, while the current local implementation is configured for Qwen2.5-7B for development. Target values cannot be interpreted as evidence of superiority until the same model and evaluation conditions are used. A final experimental paper should explicitly state hardware, temperature, decoding settings, prompt versions, and the exact test split.

A third limitation is the demonstration backend. The software includes a lightweight provider so that the architecture can be executed without a large local model. This provider should never be used to claim research-level LLM performance. Its purpose is to validate service interfaces and end-to-end orchestration.

Finally, user privacy must be considered. Conversation data can be sensitive, so production deployment should minimize data retention, use secure transport, restrict database access, and define retention policies. The prototype should store only the information needed for evaluation and demonstration.

XIII. DISCUSSION

MindBridge makes a deliberate architectural choice: it moves contextual reasoning into independently testable services. This approach has two benefits. First, it aligns with SOA learning objectives such as modularity, loose coupling, service contracts, and API-based interoperability. Second, it creates a clear

experimental structure in which each service can be evaluated individually.

The architecture also makes future extensions easier. A speech-to-text service can be added without rewriting the response service. A different emotion classifier can replace the current emotion provider. A more sophisticated safety classifier can replace the crisis service. The frontend can continue calling the same orchestrator API. Such substitutability is one of the practical strengths of service-oriented design.

However, modularity introduces orchestration cost. Each additional service can increase latency, failure probability, and debugging complexity. The latency metric is therefore not an optional detail. An architecture that produces slightly better text at a substantially higher cost may not be preferable for interactive use. Future versions should report the trade-off between response quality and latency rather than optimizing one dimension in isolation.

The project also highlights the distinction between system novelty and model novelty. The underlying LLM is not new. Multi-agent collaboration is not new. Emotional support conversation is not new. The proposed novelty lies in combining these established mechanisms with accessibility adaptation, wellness guidance, and risk-aware routing in an SOA architecture and evaluating the combined workflow systematically.

XIV. DEPLOYMENT AND SCALABILITY CONSIDERATIONS

The service decomposition is also intended to support deployment beyond a single-process prototype. A containerized implementation can package the frontend, gateway, individual services, and PostgreSQL database as independently deployable components. For demonstration purposes, a Docker Compose configuration can start the complete stack locally. For a larger deployment, the same services can be mapped to Kubernetes deployments and services, allowing the orchestration layer to scale according to traffic.

The most computationally demanding stage is expected to be response generation because it invokes the LLM. Emotion, accessibility, wellness, and crisis components may either use the same provider or use smaller specialized models. This opens an optimization path in which inexpensive classifiers handle simple routing decisions while the larger model is reserved for final response generation. Such a design could reduce latency and inference cost without changing the frontend contract.

A second scalability consideration is service parallelism. Emotion and accessibility analysis can be executed in parallel when they do not depend on each other. Wellness may depend on emotion, while the crisis service may operate independently on the conversation. A possible optimized execution graph is therefore:

$$T_{total} \approx \max(T_E, T_A, T_C) + T_W + T_R + T_{persist}, \quad (4)$$

where T_E , T_A , and T_C denote emotion, accessibility, and crisis analysis times, respectively. In contrast, strictly sequential execution would approximately sum all service times. This distinction is important because SOA decomposition should not automatically imply unnecessary sequential overhead.

Caching can also reduce repeated computation. If the same conversation state is submitted multiple times without modification, previously computed structured results can be reused. However, cache keys must not expose sensitive user content, and production deployments should consider encryption, access control, and retention policies. For a student prototype, a simple request identifier and in-memory cache may be adequate; for a production system, a dedicated cache with explicit privacy policies would be preferable.

XV. DETAILED API AND DATA CONTRACTS

Table IV summarizes the service-level contracts implemented by the proposed architecture. The contracts are intentionally small so that each service can be tested independently.

The contracts provide an explicit interface boundary between services. For example, the Response Service does not need to know whether the emotion output was produced by a rule-based development provider, a classifier, or an LLM. It only needs a schema containing an emotion label, intensity, cause, and confidence. This is a practical example of abstraction and loose coupling.

The same strategy applies to the frontend. The user interface communicates with the gateway rather than directly invoking each backend component. This prevents presentation logic from becoming dependent on service-specific implementation details. If the Crisis Service later changes its internal model, the gateway can preserve the same external response contract.

XVI. REPRODUCIBILITY AND EXPERIMENTAL PROTOCOL

Reproducibility is particularly important because the proposed comparison depends on several variables beyond the architecture. The following protocol is recommended for the final measured experiment.

First, the exact ESConv test file should be versioned and its random-selection seed recorded. Second, the model name and checkpoint should be recorded. Third, inference parameters such as temperature and decoding strategy should be fixed. Fourth, the exact prompts used by each service should be committed with the source code. Fifth, the experiment should write every generated response to a versioned result file before metric calculation. Finally, the evaluation script should load predictions from that file rather than generating responses implicitly while computing metrics.

The published MultiAgentESC experiment reports use of LLaMA3-70B and Qwen2.5-32B, 10 retrieved similar cases, an AutoGen-based implementation, temperature 0, and an A800 GPU [2]. MindBridge should record equivalent configuration details for its own run. If the same model or hardware cannot be reproduced, the final paper should describe the comparison as a reference comparison rather than a strict controlled replication.

The result state is therefore divided into three categories:

- 1) **Published:** numbers taken directly from the base paper.
- 2) **Measured:** numbers generated by running MindBridge on the specified test set with a real backend.
- 3) **Target:** desired values used to define the experiment before measurement.

This status system prevents accidental mixing of benchmark references, provisional targets, and measured findings.

A. Metric Implementation Details

The evaluation controller separates reference texts and predictions before invoking the metric functions. This design prevents a subtle experimental error in which the evaluator accidentally reads the same generated text for both sides. D-1 and D-2 are computed over generated outputs only, while BLEU, F1, and ROUGE-L compare generated outputs to the reference texts. This mirrors the metric roles described in the MultiAgentESC paper [2].

For a fair comparison, punctuation normalization, tokenization, case handling, and sentence splitting should be kept identical across systems. The evaluation environment should also record library versions. Although these choices may appear minor, metric implementations can vary across libraries and change reported values. Reproducibility therefore depends on recording the exact evaluation software as well as the model.

B. Human Evaluation Extension

Automatic metrics cannot determine whether a supportive answer feels appropriate to a user. MultiAgentESC addresses this limitation using blinded human comparisons across fluency, identification, comforting, suggestion, and overall support [2]. MindBridge can adopt the same dimensions while adding accessibility-specific questions. For example, evaluators can judge whether the response respects the declared communication preference and whether the language complexity is appropriate for the scenario.

A future study can recruit multiple evaluators and randomize response order. Inter-rater agreement should be measured, and evaluators should be blinded to the identity of the generating system. This would provide stronger evidence than presenting only automatic overlap metrics.

XVII. THREATS TO VALIDITY

Several threats to validity must be considered. The first is benchmark mismatch. ESConv was created for emotional-support conversation and contains supporter responses, but it does not fully represent all accessibility conditions or crisis scenarios envisioned by MindBridge. Therefore, success on ESConv should not be interpreted as proof that the complete platform works equally well for every disability group.

The second threat is metric mismatch. High lexical overlap with a reference can be achieved by producing a superficially similar response, while a novel but appropriate response may have lower BLEU or ROUGE-L. Conversely, high D-1 or D-2 can reflect novelty without usefulness. The final study should therefore treat automatic metrics as one component of the evidence.

The third threat is model dependence. The published base values use Qwen2.5-32B. A MindBridge run with a different model size or provider may change performance independently of the architectural modification. This is why the final measured

TABLE IV
REPRESENTATIVE MINDBRIDGE SERVICE CONTRACTS

Service	Endpoint	Input	Output
Emotion	/emotion/analyze	conversation text	emotion, intensity, cause, confidence
Accessibility	/accessibility/analyze	conversation and user preferences	needs, language level, preferred mode, adaptation
Wellness	/wellness/recommend	conversation and emotion structure	goal and general wellness suggestions
Crisis	/crisis/detect	conversation text	risk level, crisis flag, evidence, confidence
Response	/response/generate	conversation plus all service outputs	final supportive response
Orchestrator	/api/mindbridge/analyze	conversation and preferences	all service outputs plus final response and latency
Evaluation	/evaluation/run	experiment configuration	metrics, comparison table, status

comparison should use the same model whenever resources permit.

The fourth threat is prompt sensitivity. Multi-agent systems can be strongly affected by system instructions, output schemas, context ordering, and temperature. Prompt versions should therefore be fixed during an experiment, and an additional sensitivity analysis can be conducted later.

The fifth threat concerns safety evaluation. Crisis detection is a high-stakes task. A small synthetic or manually constructed test set cannot establish safe real-world performance. Before deployment, the crisis component would require substantially broader validation, explicit safety review, and carefully designed escalation pathways.

The final threat is ecological validity. Users may communicate differently in real deployments than in curated datasets. They may use code-switching, slang, ambiguous language, incomplete messages, speech recognition errors, or culturally specific expressions. Future testing should therefore include diverse interaction modes and user-centered evaluation rather than relying exclusively on benchmark data.

APPENDIX

The following example illustrates how a single request moves through the MindBridge service graph. Consider the input: “I am stressed about my exams and I have stopped talking to my friends.” The gateway first normalizes the request and passes the conversation to the specialized services. A development-mode trace can be represented as follows:

```

USER MESSAGE
|
+--> Emotion Service
|     emotion = anxiety
|     intensity = 0.70
|     confidence = 0.80
|
+--> Accessibility Service
|     preferred_mode = text
|     language_level = normal
|
+--> Wellness Service
|     goal = support
|     suggestions = [social connection
                    , small routine]

```

```

|
+--> Crisis Service
|     risk_level = low
|     crisis_detected = false
|
+--> Response Service
|     final supportive response

```

The values shown above describe the development demonstration path and are not benchmark measurements. Their purpose is to demonstrate the type of structured information exchanged between services. In a real experiment, the emotion, wellness, and crisis outputs would be produced by the configured model backend and preserved alongside the final response for auditability.

An example orchestrator request can be expressed as:

```

POST /api/mindbridge/analyze
{
  "conversation": "I am stressed about exams...",
  "user_preferences": {
    "mode": "text",
    "language_level": "normal"
  }
}

```

The corresponding structured response can be represented as:

```

{
  "emotion": {...},
  "accessibility": {...},
  "wellness": {...},
  "crisis": {...},
  "response": "...",
  "latency_ms": 0
}

```

This data-centric representation is useful for both SOA testing and research evaluation. A researcher can inspect each intermediate result, reproduce a failure, and compare service-level behavior without reverse engineering the final response. The same trace can also be stored in PostgreSQL for later analysis, provided that privacy controls are applied.

The project can further expose the trace through a development-only dashboard. Such a dashboard should clearly

distinguish internal reasoning metadata from user-facing content and should not expose sensitive intermediate information in a production setting. The research value of the trace is primarily reproducibility: it allows the authors to explain why a response was produced and which service supplied the contextual signal.

The appendix also illustrates why the proposed architecture should not be reduced to a single prompt. Even when a single LLM ultimately generates the final response, the system around the model defines structured contracts, safety routing, accessibility adaptation, data persistence, and evaluation. These components are the primary SOA contribution of the prototype.

Several extensions are planned. First, the evaluation should be reproduced with Qwen2.5-32B under settings matching the base paper, enabling a fair direct comparison. Second, a human evaluation should be conducted using blinded response pairs and the dimensions of fluency, identification, comforting, suggestion, and overall support. Third, the crisis component should be tested using a separately labelled risk dataset and reported with precision, recall, F1, false-positive rate, and false-negative rate rather than a single informal reliability score.

Fourth, accessibility should be evaluated with task-based scenarios covering visual, hearing, speech, physical, and cognitive requirements. Fifth, the project can be extended to voice interaction using speech recognition and text-to-speech. Sixth, user-level personalization can be added while preserving privacy and explicit consent. Seventh, the service layer can be containerized and deployed using Kubernetes so that scaling characteristics can be studied empirically.

A final research direction is multi-objective optimization. Instead of optimizing only response metrics, the system could jointly consider response quality, diversity, safety detection, accessibility conformance, and latency. Such a study would better reflect the real engineering problem faced by an accessible mental-wellness platform.

This paper presented MindBridge AI, a service-oriented multi-agent platform for accessible and context-aware digital mental-wellness support. The architecture decomposes the interaction workflow into emotion, accessibility, wellness, crisis-risk, and response services coordinated through a central orchestrator. The design follows SOA principles of loose coupling, modularity, reusable interfaces, and independent service testing.

The work is grounded in the ESConv benchmark and uses MultiAgentESC as the primary research baseline. The same automatic metrics—Distinct-1, Distinct-2, BLEU-1/2/3, F1, and ROUGE-L—are incorporated into the evaluation pipeline. The base-paper Qwen2.5-32B results provide a fixed numerical reference. MindBridge defines a positive provisional target profile above these values, motivated by the additional contextual signals available to the final response generator. These target values are explicitly not presented as measured findings; the next research step is a controlled full-LLM experiment on the 100-conversation test set.

More broadly, MindBridge demonstrates how an SOA perspective can be combined with LLM-based multi-agent interaction to create a modular academic prototype. The

architecture provides a practical foundation for future work on accessibility, safety, multimodal interaction, human evaluation, and reproducible benchmarking while maintaining clear boundaries between wellness assistance and clinical care.

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